

Fig. 2: Actual-size PCB layout for the bipolar power supply

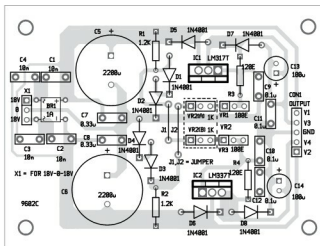


Fig. 3: Components layout for the PCB

D8) and a few other components.

The 230V AC mains is fed to the primary coil of transformer X1. You can choose the transformer as per your maximal output voltage and current requirements. Here, transformer X1 is used to produce regulated output voltage up to $\pm 15V$.

Bridge rectifier BR1 should be rated at least 1A. Main filtering capacitors C5 and C6 should be at least 2200 μF , 40V. The unregulated positive voltage is applied to pin 3 of IC1, while the unregulated negative voltage is applied to pin 2 of IC2.

Regulated power supply section includes LM317T, LM337T and stereo potmeter VR2(A) + VR2(B) for simultaneous adjustment of the output voltages. The output voltage of LM317T normally starts from around 1.25V and that of LM337T from around -1.25V. Here, D1 and D2 create positive reference voltage of around +1.3V, which is used as the offset for IC2. Also, D3 and D4 create negative reference voltage of around -1.3V, which is used as the offset for IC1.

That is why output voltages V3 and V4 can start from almost ground levels. If you need better stability, use reference diodes for 1.2V, such as LM385-1.2, instead of ordinary diodes D1 through D4. Diodes D5 through D8 protect regulators from

reverse voltages.

Heat-sinks. Mount IC1 and IC2 on appropriate heat-sinks that have thermal resistance value of less than 4°C/W. The maximal power dissipation can go up to 10W if you need output current above 0.5A at the lowest output voltages. When calculating the required heat-sink size, consider that the maximal power dissipation of LM317T and LM337T in TO-220 package is 20W, junction-to-case thermal resistance is 4°C/W and maximal junction temperature is +125°C.

Construction and testing

An actual-size PCB layout for the bipolar power supply is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect the secondary terminals of transformer to points marked X1 in the PCB. Fix potmeters VR1 through VR3 on the front side of the cabinet so that you can adjust the voltages easily.

For testing, connect the circuit to 230V AC mains. Next, at the output connector, connect load resistors of 33 to 51 ohms with power dissipation of at least 10W (preferably above 20W). Set wipers of VR1 and VR3 to their middle positions. By varying stereo potmeter VR2(A) + VR(B), adjust the output voltage to the required voltages,

PARTS LIST

Semiconductors:

- IC1 - LM317T positive-voltage regulator
- IC2 - LM337T negative-voltage regulator
- BR1 - 1A bridge rectifier
- D1-D8 - 1N4001 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 1.2-k Ω -ohm
- R3, R4 - 120-ohm
- VR1, VR3 - 100-ohm potmeter
- VR2 - 1-k Ω -ohm stereo potmeter

Capacitors:

- C1-C4 - 10nF ceramic disk
- C5, C6 - 2200 μF , 40V electrolytic
- C7, C8 - 0.33 μF ceramic disk
- C9-C12 - 0.1 μF ceramic disk
- C13, C14 - 100 μF , 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 18V-0-18V, 1A secondary transformer
- CON1 - 5-pin connector
- Heat-sink for LM317T and LM337T

e.g., around $\pm 10V$. Vary VR1 and/or VR3 in case further adjustment is required in the output voltage. Now if you connect and disconnect the loads, output voltages V3 and V4 should change slightly around the initial value of $\pm 10V$. **EFY**



Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFPPT(Casablanca, Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria